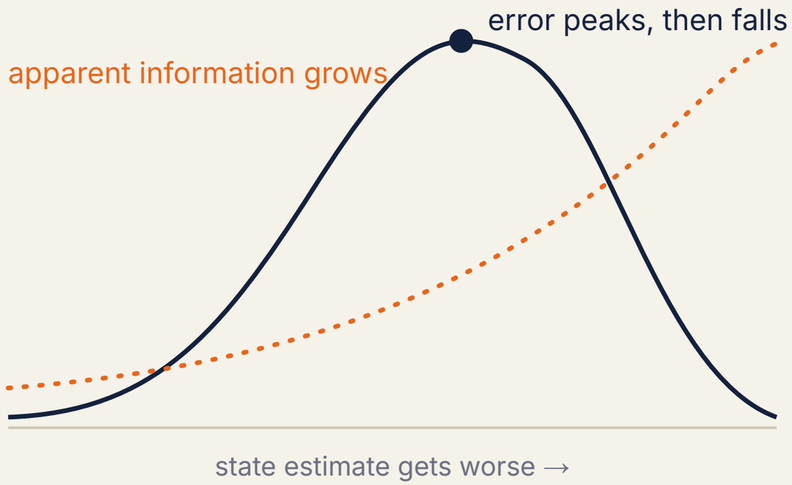
MINOR PROJECT · AISTATS 2027 SUBMISSION

Reliable causal estimation under latent *Markov* confounding



Overlap-aware regime Double Machine Learning, tested on air-quality sensors in four Indian cities

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Guide: Dr. Savitha G · School of Computer Engineering · MIT Manipal · November 2026

THE PROJECT IN ONE SLIDE

A trust rule everyone uses *fails*. We show exactly how, and fix the estimator.

4,900

simulation runs matched by our closed-form law, within 2%

82%

of learned-state runs where a wrong state looks better conditioned than the true one

4

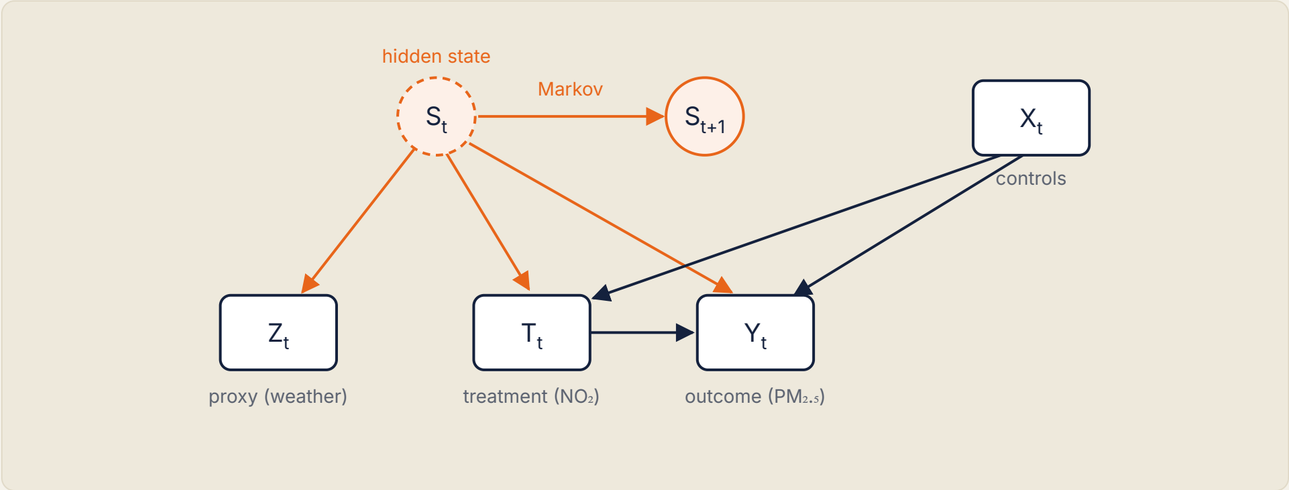
Indian cities: Delhi, Mumbai, Bengaluru, Kolkata

THE QUESTION

Does cutting NO_2 cut $PM_{2.5}$?

A forecast cannot answer this. The atmosphere has a hidden state, such as stagnation or ventilation, that raises both pollutants together.

We never observe that state. We only see weather proxies.



OBJECTIVES

Six things we set out to do.

01 Assemble and audit a multi-city hourly dataset

02 Characterize how state error becomes causal error

03 Build a reliable regime-aware estimator

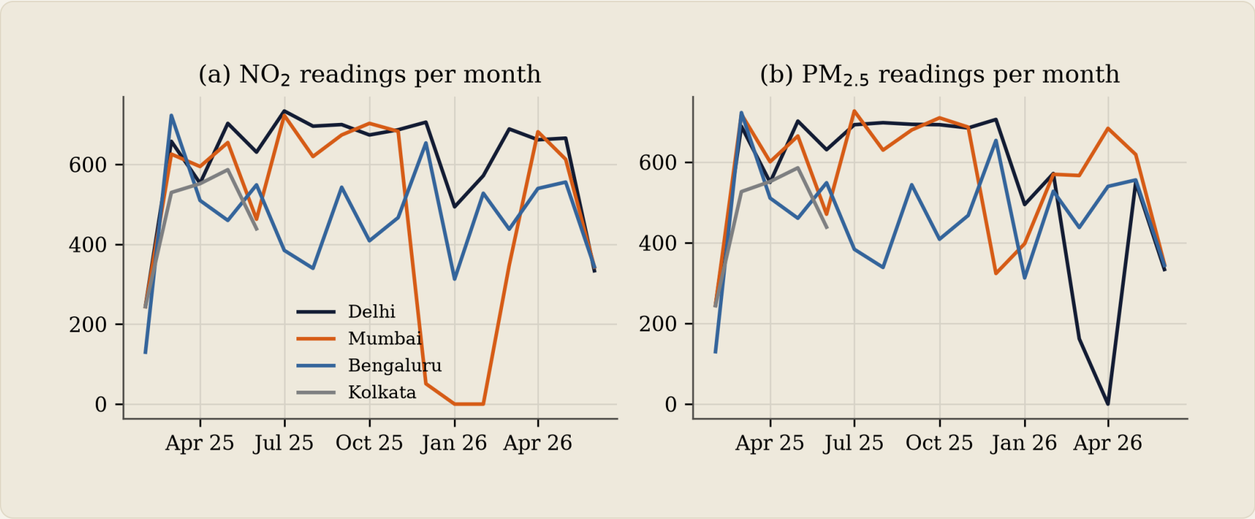
04 Validate from synthetic to real data

05 Produce a peer-reviewed research submission

06 Make every number reproducible

DATA

Sixteen months of hourly *sensor* data.



OpenAQ / CPCB pollution and Open-Meteo weather, February 2025 to June 2026.

A rounding bug had put “hourly” data on a two-hour grid. We found it and recomputed everything.

WHY STANDARD METHODS FAIL

Standard DML is off by *5×*, at every proxy quality.

8.44

bias of standard Double ML

1.53

true average effect in the benchmark

Adding more observed controls can make the bias worse, not better.

APPROACH

Infer the state, then estimate *carefully*.



posterior enters twice: adjustment feature (fine) and score weight (coarse target)

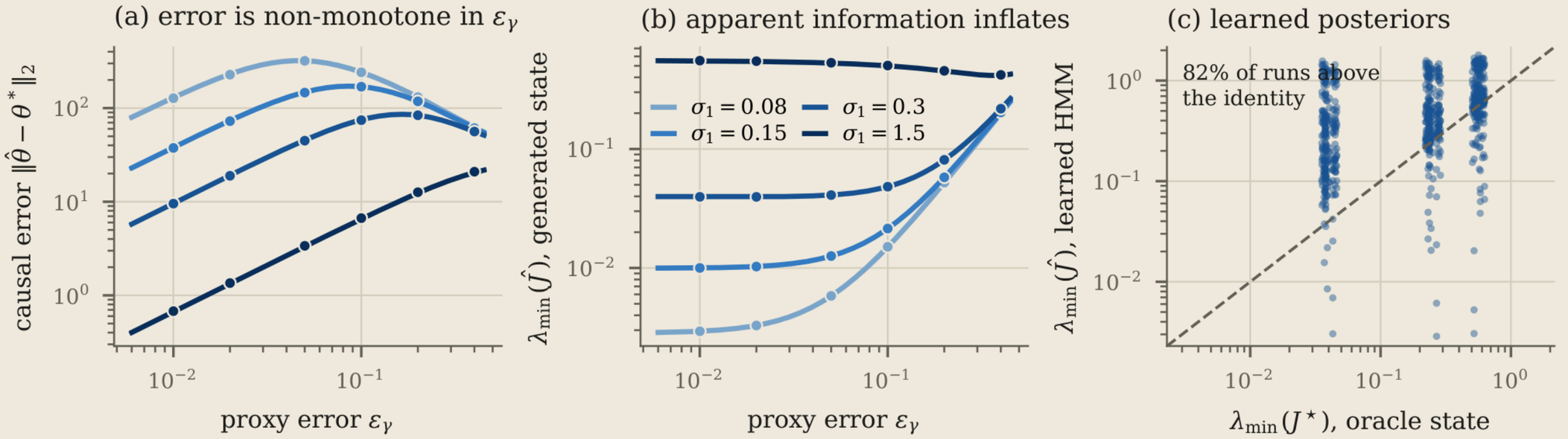
Hidden Markov model gives a probability of each regime at each hour.

Cross-fitting skips neighbouring hours so time dependence cannot leak.

A regularized coupled score gives one effect per regime.

KEY DISCOVERY

A worse state estimate can look *better*.



We call it **phantom causal resolution**: error rises while the usual conditioning check improves, up to 70×.

THE RULES: WHY IT FAILS

Two *laws* explain the failure.

Omitted-regime bias

$$\hat{\theta}_{\text{naive}} \to \sum_k \omega_k \theta_k^* + \mathcal{B}_{\text{het}} + \mathcal{B}_{\text{conf}}$$

Controls alone cannot see the hidden regime. More controls can make the bias worse.

State-confusion law

$$\lambda_- = \tfrac{1}{2}\sigma^2(1-\varepsilon_\gamma)^2 + \tfrac{1}{8}\varepsilon_\gamma^2\Delta^2$$

Conditioning improves as the state estimate gets worse. Matches 4,900 runs.

THE RULES: WHAT WE BUILD FROM THEM

Three *rules* shape the estimator.

Orthogonality

$$\partial_{(m,\mu)}\, \mathbb{E}\, \psi_k = 0$$

Condition nuisances on the posterior: first-order errors drop out for every posterior.

Error bound

$$\|\hat{\theta} - \theta^*\| \lesssim \frac{\varepsilon_\gamma}{\lambda_{\min}(J)}$$

State error and nuisance error add; neither hides the other.

Residual-state bias

$$B = \frac{a\, b\, v}{a^2 v + \sigma^2}$$

State error × treatment shift × outcome shift, over a variance the shift inflates.

THE ESTIMATOR

Condition on the posterior. *Don't* weight by it.

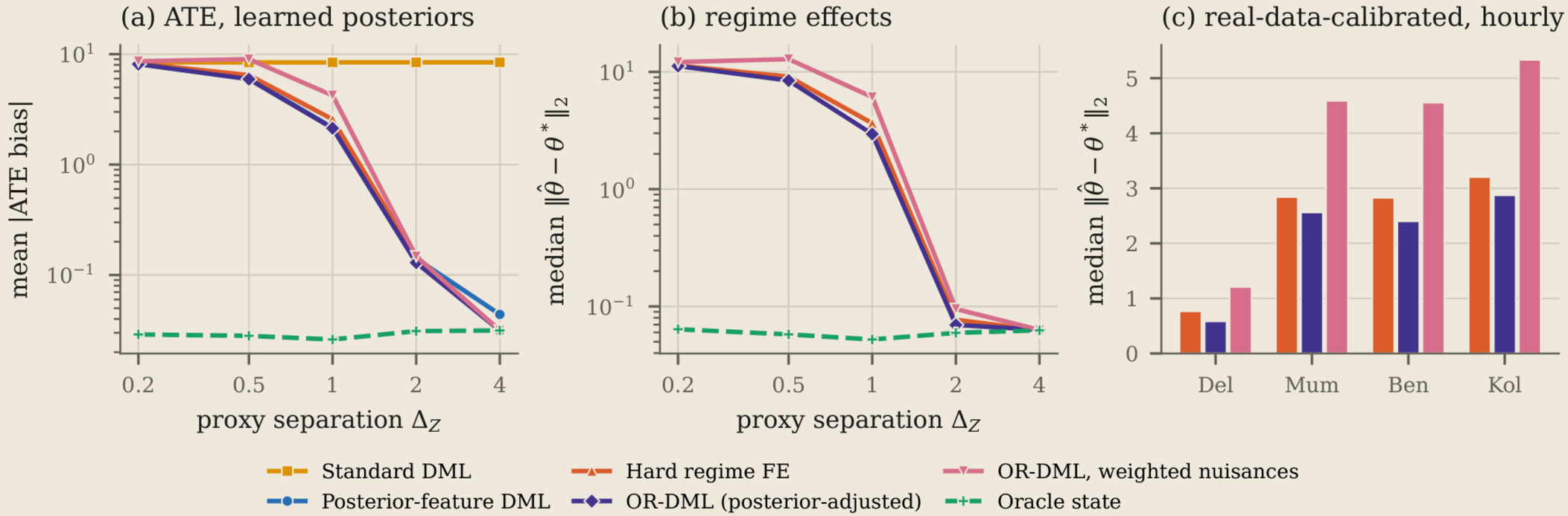
```
F = np.column_stack([X, G] + [G[:, [k]] * X for k in range(K-1)])
for train, test in purged_blocks(N, embargo):
    m_y = fit(F[train], Y[train])      # posterior as a feature
    m_t = fit(F[train], T[train])
J = weighted_gram(tilde_T, w)           # posterior as a weight
theta = inv(J + lam * I) @ S
```

85% → 0.4%

non-orthogonality of the old weighted design versus ours

RESULTS

Best in *every* city.



Purple: ours. Pink: our earlier design. Below separation of about 1, every method fails.

REAL SENSORS, HONEST VERDICTS

What the four cities *can* say.

CITY	VERDICT	WHY
Delhi	ASSOCIATION	Positive in both regimes. Needs an 82 SD hidden shift to vanish, but a lead placebo rejects: not an emissions effect.
Mumbai	INCONCLUSIVE	One 277-hour constant NO_2 run; screening removes 23% of the data.
Bengaluru	INCONCLUSIVE	Fails placebo checks.
Kolkata	INCONCLUSIVE	Only 16 distinct NO_2 values; screened effect fails placebo.

REPRODUCIBILITY

Every number traces to a *file*.

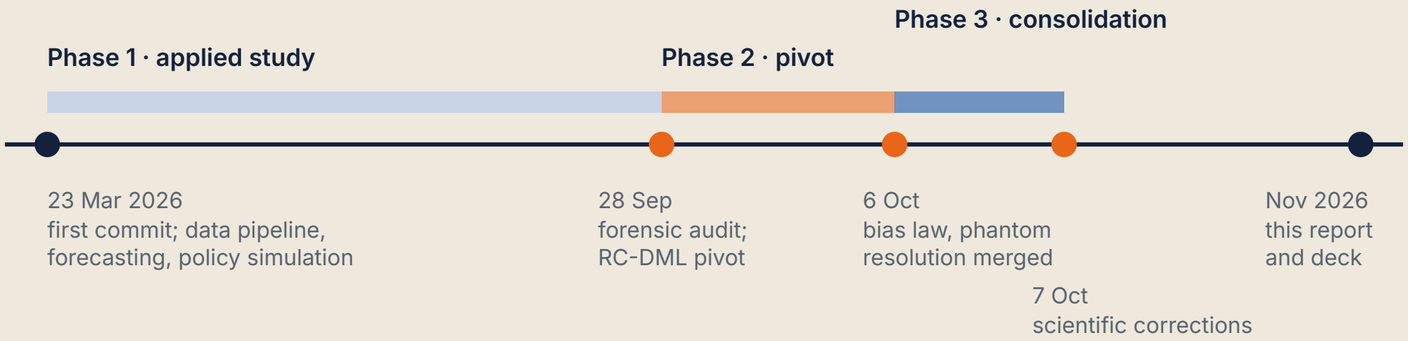
```
macro,value,source,how
FacRhoD,0.995,reports/factorial_reliability_correlations.csv,Spearman_Run_Difficulty
LPFracInfl,82,reports/synthesis/learned_posterior_coupled_meta.json,% runs lambda_proxy>lambda_oracle
...                                  # 247 rows in claims.csv
```

```
python verify_submission.py --fast
bash reproduce.sh        # ~12 min on 2 cores
```

One script writes every table and number in the paper. A verifier fails on any mismatch.

THE JOURNEY

We audited our own work, and *retired* claims.



OUT A health-benefit figure built on an arbitrary scalar

OUT Lags that were “hourly” but two hours

OUT Placebo tests that were mechanically zero

WHERE IT IS SUBMITTED

A top-tier, *peer-reviewed* venue.

A

CORE conference rank (A* flagship, A excellent)

31%

of submissions accepted in 2025: 583 of 1,861

1985

series began, proceedings published in PMLR

AISTATS 2027 sits where machine learning meets statistics. In AI, conferences like this are full, blind-reviewed publications, not just talks.

CONCLUSIONS

Two rules *survive*.

Design

Use the posterior as an adjustment feature. Use it as a weight only on the target.

Reporting

State the leftover state variance, the regime shifts, a robustness value, simple baselines, and aligned placebos.

LIMITS AND NEXT STEPS

What we *do not* claim.

Limits

Every method fails at weak proxy separation. City results are associations. Inference theory excludes full-sample smoothing.

Next

Open-source package. Policy-shock designs for emission claims. Re-test the dual-factor idea of ambiguity times conditioning.

THANK YOU

Questions?

Pulkit Kumar · Utsav Verma · Nishant Kumar Upadhyay

Guide: Dr. Savitha G · School of Computer Engineering · MIT Manipal

Report, manuscript, code and reproduction pipeline are in the project repository.